



cpu-E5-2680v2-ubuntu-24.04

KVM testing on Ubuntu 24.04 via the Phoronix Test Suite.

Test Systems:

Ubuntu 24.04

Processor: Intel Xeon E5-2680 v2 (8 Cores), Motherboard: QEMU Standard PC (Q35 + ICH9 2009) (4.2025.05-2 BIOS), Chipset: Intel 82G33/G31/P35/P31 + ICH9, Memory: 1 x 16 GB RAM QEMU, Disk: 32GB QEMU HDD, Audio: Intel 82801I, Network: Red Hat Virtio device

OS: Ubuntu 24.04, Kernel: 6.8.0-101-generic (x86_64), Compiler: GCC 13.3.0, File-System: ext4, System Layer: KVM

Kernel Notes: Transparent Huge Pages: madvise

Compiler Notes: --build=x86_64-linux-gnu --disable-vtable-verify --disable-werror --enable-bootstrap --enable-cet --enable-checking=release --enable-clocale=gnu
--enable-default-pie --enable-gnu-unique-object --enable-languages=c,ada,c++,go,d,fortran,objc,obj-c++,m2 --enable-libphobos-checking=release
--enable-libstdcxx-backtrace --enable-libstdcxx-debug --enable-libstdcxx-time=yes --enable-link-serialization=2 --enable-multiarch --enable-multilib --enable-nls
--enable-objc-gc=auto --enable-offload-defaulted
--enable-offload-targets=nvptx-none=/build/gcc-13-EldibY/gcc-13-13.3.0/debian/tmp-nvptx/usr,amdgc-n-amdhsa=/build/gcc-13-EldibY/gcc-13-13.3.0/debian/tmp-gcn/usr
--enable-plugin --enable-shared --enable-threads=posix --host=x86_64-linux-gnu --program-prefix=x86_64-linux-gnu- --target=x86_64-linux-gnu --with-abi=m64
--with-arch-32=i686 --with-build-config=bootstrap-lto-lean --with-default-libstdcxx-abi=new --with-gcc-major-version-only --with-multilib-list=m32,m64,mx32

--with-target-system-zlib=auto --with-tune=generic --without-cuda-driver -v

Processor Notes: CPU Microcode: 0x42d

Security Notes: gather_data_sampling: Not affected + indirect_target_selection: Not affected + itlb_multihit: Not affected + l1tf: Mitigation of PTE Inversion; VMX: flush not necessary SMT disabled + mds: Vulnerable: Clear buffers attempted no microcode; SMT Host state unknown + meltdown: Mitigation of PTI + mmio_stale_data: Unknown: No mitigations + reg_file_data_sampling: Not affected + retbleed: Not affected + spec_rstack_overflow: Not affected + spec_store_bypass: Mitigation of SSB disabled via prctl + spectre_v1: Mitigation of usercopy/swaps barriers and __user pointer sanitization + spectre_v2: Mitigation of Retpolines; IBPB: conditional; IBRS_FW; STIBP: disabled; RSB filling; PBRSE-IBRS: Not affected; BHI: Retpoline + srbds: Not affected + tsa: Not affected + tsx_async_abort: Not affected + vmscape: Not affected

Ubuntu 24.04

7-Zip Compression - Compression Rating (MIPS) 22614

Standard Deviation 11.8%

7-Zip Compression - D.R (MIPS) 19729

Standard Deviation 11.4%

C-Ray - 4K - 16 (sec) 854.887

Standard Deviation 5.5%

C-Ray - 5K - 16 (sec) 1464

Standard Deviation 3.9%

C-Ray - 1080p - 16 (sec) 212.075

Standard Deviation 4.6%

OpenSSL - SHA256 (byte/s) 1173206457

Standard Deviation 8.3%

OpenSSL - SHA512 (byte/s) 1326465780

Standard Deviation 8.1%

OpenSSL - RSA4096 (sign/s) 748.4

Standard Deviation 5.8%

OpenSSL - RSA4096 (verify/s) 48238

Standard Deviation 4.7%

OpenSSL - ChaCha20 (byte/s) 8582917143

Standard Deviation 4.3%

OpenSSL - AES-128-GCM (byte/s) 6891612567

Standard Deviation 3.5%

OpenSSL - AES-256-GCM (byte/s) 6684505520

Standard Deviation 1.8%

OpenSSL - ChaCha20-Poly1305 (byte/s) 5885552930

Standard Deviation 3.1%

Stockfish - Chess Benchmark (Nodes/s) 2943158

Standard Deviation 7.6%

7-Zip Compression 25.00

Test: Compression Rating



1. (CXX) g++ options: -lpthread -ldl -O2 -fPIC

7-Zip Compression 25.00

Test: Decompression Rating



1. (CXX) g++ options: -lpthread -ldl -O2 -fPIC

C-Ray 2.0

Resolution: 4K - Rays Per Pixel: 16



1. (CC) gcc options: -lpthread -lm

C-Ray 2.0

Resolution: 5K - Rays Per Pixel: 16



1. (CC) gcc options: -lpthread -lm

C-Ray 2.0

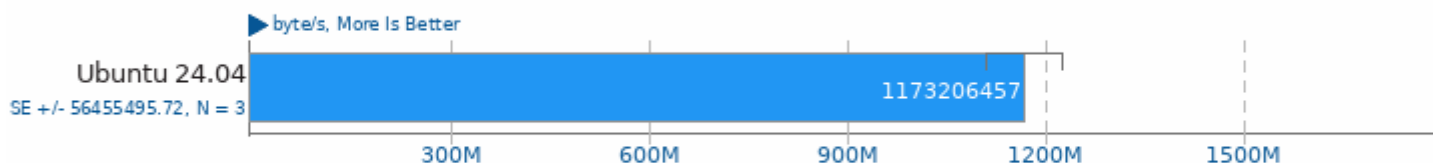
Resolution: 1080p - Rays Per Pixel: 16



1. (CC) gcc options: -lpthread -lm

OpenSSL 3.6

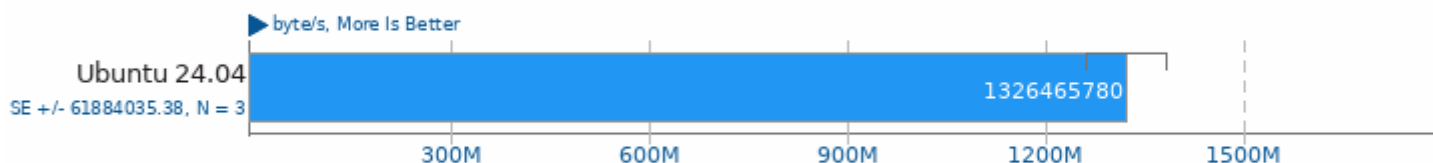
Algorithm: SHA256



1. (CC) gcc options: -pthread -m64 -O3 -lssl -lcrypto -ldl

OpenSSL 3.6

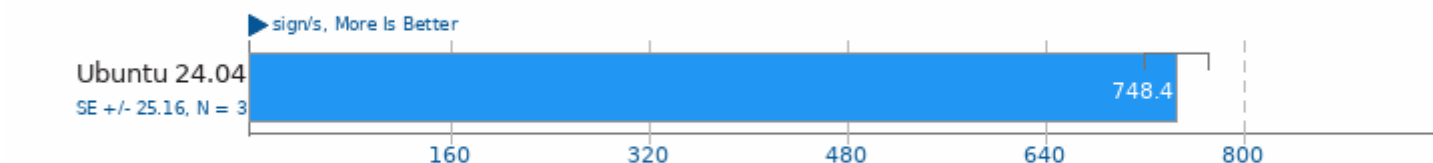
Algorithm: SHA512



1. (CC) gcc options: -pthread -m64 -O3 -lssl -lcrypto -ldl

OpenSSL 3.6

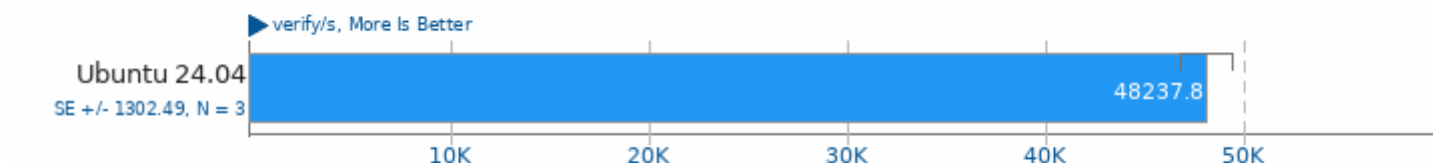
Algorithm: RSA4096



1. (CC) gcc options: -pthread -m64 -O3 -lssl -lcrypto -ldl

OpenSSL 3.6

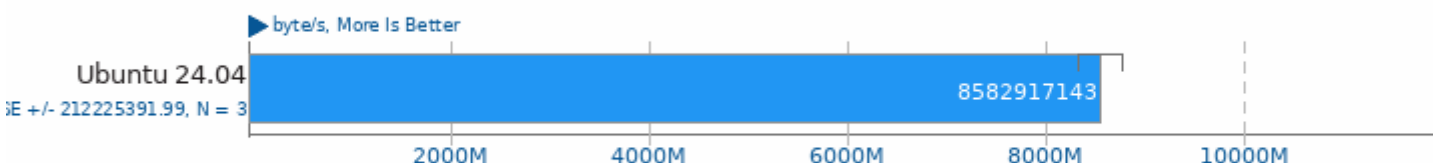
Algorithm: RSA4096



1. (CC) gcc options: -pthread -m64 -O3 -lssl -lcrypto -ldl

OpenSSL 3.6

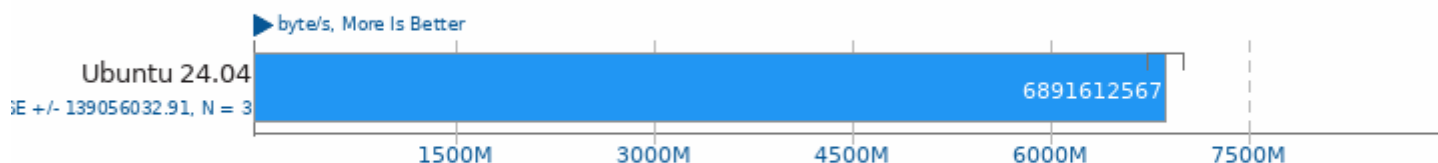
Algorithm: ChaCha20



1. (CC) gcc options: -pthread -m64 -O3 -lssl -lcrypto -ldl

OpenSSL 3.6

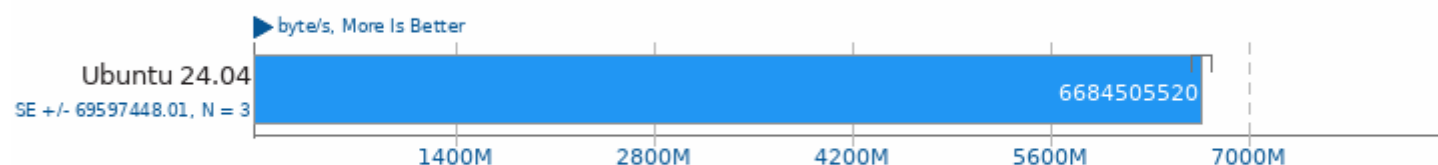
Algorithm: AES-128-GCM



1. (CC) gcc options: -pthread -m64 -O3 -lssl -lcrypto -ldl

OpenSSL 3.6

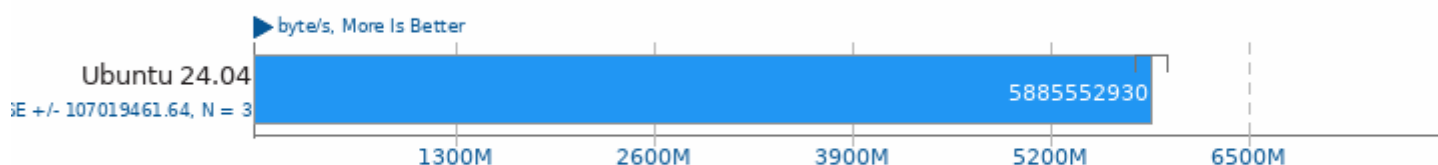
Algorithm: AES-256-GCM



1. (CC) gcc options: -pthread -m64 -O3 -lssl -lcrypto -ldl

OpenSSL 3.6

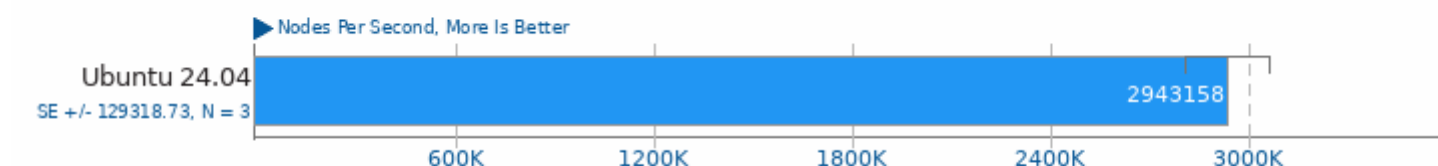
Algorithm: ChaCha20-Poly1305



1. (CC) gcc options: -pthread -m64 -O3 -lssl -lcrypto -ldl

Stockfish 17.1

Chess Benchmark



1. (CXX) g++ options: -lgcov -m64 -lpthread -fno-exceptions -std=c++17 -fno-peel-loops -fno-tracer -pedantic -O3 -funroll-loops -msse -msse3 -mpopcnt -msse4.1 -r

This file was automatically generated via the Phoronix Test Suite benchmarking software on Saturday, 14 March 2026 14:34.